

HARIKA PRATHIPATI

Python ML Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer with 3+ years of production experience building and deploying ML systems in healthcare environments. Recently completed a Master's in Artificial Intelligence (specialization: MLOps, Cloud AI Systems) and transitioned to an ML engineering role. Strong foundation in end-to-end ML pipelines, model optimization, and HIPAA-compliant data workflows. Experienced in feature engineering, model monitoring, and production deployment. Seeking ML Engineer or AI Engineer roles in healthcare, enterprise, or GenAI domains.

TECHNICAL SKILLS

Programming Languages: Python (Advanced), SQL (Advanced), PySpark, Bash Scripting

Machine Learning & AI: Scikit-Learn, XGBoost, Random Forest, Logistic Regression, Gradient Boosting, Ensemble Methods, Cross-Validation, Hyperparameter Tuning, SHAP Analysis, Model Evaluation

Generative AI & LLMs (Foundational): LLM fundamentals, transformer architecture concepts, embeddings, prompt engineering basics, RAG concepts, NLP pipelines

Model Development: Feature Engineering, Feature Selection, Class Imbalance Handling (SMOTE, Class Weights), Model Evaluation (Precision, Recall, F1-Score, AUC-ROC), Bias-Variance Analysis

MLOps & Production: MLflow (Experiment Tracking, Model Registry), Model Versioning, Model Monitoring, Data Drift Detection, Docker, FastAPI, Flask, Model Serialization

Data Engineering: Apache Spark, PySpark, ETL Pipeline Development, Apache Airflow, Data Warehousing, SQL Query Optimization, Data Quality Validation

Cloud Platforms: AWS (EC2, S3, Lambda, CloudWatch, Glue, SageMaker) – Production experience, Azure (Azure AI Services, Azure Data Lake, Azure ML) – Certified proficiency

Data Preprocessing: Missing Data Imputation, Feature Scaling (StandardScaler, MinMaxScaler), Categorical Encoding (One-Hot, Label, Target), Data Validation

Databases: PostgreSQL, MySQL, MongoDB, Redis, Database Design, Query Optimization

DevOps & Testing: Git/GitHub, Unit Testing (PyTest, unittest), Code Review, Logging, Debugging, Documentation

PROFESSIONAL EXPERIENCE

Python ML Engineer

July 2025 – Present

Maximus | Texas, United States

- Developed and deployed classification models (Logistic Regression, Random Forest, XGBoost) for patient risk stratification and hospital readmission prediction using FastAPI REST API architecture, achieving strong performance on hold-out test datasets.
- Engineered clinical feature sets including temporal patterns, medication interactions, comorbidity indices, and lab trends; improved model F1-score compared to baseline implementations through systematic feature creation.
- Built production ML pipelines processing large-scale patient records using Python, PySpark, and SQL; optimized Spark transformations to reduce data processing time through efficient partitioning and caching strategies.
- Implemented MLOps workflows using MLflow for experiment tracking and model versioning; deployed containerized models on AWS Lambda with monitoring via CloudWatch for tracking prediction latency and error rates.
- Developed automated data validation scripts and Airflow workflows to detect schema violations and missing values before training; created reusable Python modules for feature engineering and preprocessing.

- Collaborated with data engineers, clinical SMEs, product managers, and software engineers; presented model performance and technical findings to non-technical stakeholders.
- Worked within HIPAA compliance frameworks, ensuring data anonymization, encryption at-rest and in-transit, and role-based access controls for patient data processing workflows.
- Optimized SQL queries with multi-table joins and window functions for data extraction pipelines, improving performance for model training dataset generation.
- Conducted hyperparameter optimization across model architectures using GridSearchCV and RandomizedSearchCV; addressed class imbalance using SMOTE and class weight adjustments to optimize precision-recall tradeoff.

Python ML Engineer
Wipro Limited | Hyderabad, India

August 2021 – July 2023

- Built Apache Spark (PySpark) data transformation pipelines processing structured and semi-structured healthcare data from multiple source systems into a centralized data warehouse; implemented data quality checks and validation rules.
- Developed Python-based ETL modules for Medicaid and Medicare analytics platforms processing large beneficiary record datasets with high data completeness; ensured data integrity across pipeline stages.
- Implemented machine learning proof-of-concept studies using Scikit-Learn for healthcare classification tasks, establishing baseline models to validate ML feasibility for stakeholder decision-making.
- Created Python automation scripts for data validation and quality checks, reducing manual validation effort and enabling the team to focus on higher-value analytical initiatives.
- Performed feature engineering and preprocessing on healthcare datasets, including missing value imputation, categorical encoding, and feature scaling to support downstream modeling efforts.
- Participated in code reviews with senior developers, ensuring adherence to coding standards and best practices; contributed to pull request reviews and knowledge sharing sessions.
- Optimized complex SQL queries with multi-table joins and subqueries to reduce query execution time for business intelligence and reporting workflows.
- Maintained Git version control best practices with feature branching and comprehensive documentation; supported knowledge transfer across distributed teams.
- Supported REST API integration between healthcare applications and external partner systems; debugged data exchange issues to ensure reliable system interoperability.
- Built data visualization dashboards using Matplotlib and Seaborn to communicate analytical insights and data quality metrics to business stakeholders.
- Participated in requirements gathering sessions, translating business needs into technical specifications for data processing features and analytics capabilities.
- Documented data pipeline architectures, ETL processes, and data models to support team onboarding; created technical guides for common workflows and troubleshooting.

EDUCATION

Master of Science in Artificial Intelligence

May 2025

University of North Texas | Denton, TX

Bachelor of Technology in Electronics and Communication Engineering

July 2021

KKR & KSR Institute of Technology and Sciences | Andhra Pradesh, India

CERTIFICATIONS & CREDENTIALS

- Microsoft Azure AI Fundamentals
- Microsoft Azure AI Engineer Associate
- NVIDIA Certified Associate – Generative AI & LLMs (NCA-GENL)